

Plasticity → SVAMP Pilot — Three-Run Summary

Qwen2.5-0.5B-Instruct · GRPO on GSM8K (stage A) · SVAMP learnability probe (stage C) · single seed each

Bottom line: Across three learning rates, standard KL-free GRPO on this setup reliably collapses under sustained training — before a clean “trains-well-while-capacity-erodes” regime ever appears. The hypothesis is neither confirmed nor cleanly refuted: the decisive test regime hasn’t been reached, and reaching it likely needs a design change (a KL anchor), not another LR.

What we were testing

The proposal's headline claim is that a plasticity metric Q (effective rank, dormant-neuron fraction), read during stage A, warns that a later stage will stall *earlier* than the signals trainers already watch (reward plateau, KL, gradient norm). The pilot's job was a first, cheap check: train a 0.5B model on GSM8K, and from checkpoints along the way measure whether Q moves and whether it tracks the model's ability to then learn SVAMP.

The three runs at a glance

	Run 1	Run 2	Run 3
Stage-A LR	5e-6, cosine→0	1e-5, constant	5e-6, constant
KL (β)	0	0	0
Updates planned / actual	200 / 200	450 / 450	450 / 110 (aborted)
Intended regime	gentle	hot	medium
Stage A outcome	trained, mild drift	destroyed by ~ckpt 50	collapsed ~step 90, guard fired
GSM8K acc (start→end)	≈0.45 → 0.39	0.45 → 0	0.45 → 0.23 (@ckpt 50)
Effective rank	flat ($\pm 1\%$)	collapsed 1708→492 (ckpt 150→300)	flat over surviving ckpts
Dormant fraction	0 everywhere	0 → 0.0002 (trivial)	0 everywhere
Entropy behaviour	0.37→0.10 by ~upd 90	spiked 0.59 then crashed	0.32→0.16, then collapse
SVAMP improvement	+0.06/+0.07 then −0.05/−0.07*	+0.12/+0.17 then 0/0/0	+0.16/+0.19 (both survivors)
Metric vs. dashboard	nothing moved	Q lagged failure 100+ upd	collapsed before metrics
Verdict	too cold	too hot	still collapses

* Run 1's negative SVAMP improvements were later traced partly to a greedy-eval truncation artifact (fixed in runs 2–3), so their magnitude is not trustworthy.

Reading across the runs

The runs bracket the target regime from three sides and miss it each time. Run 1 was too gentle — the model trained fine but nothing eroded, so Q had nothing to detect. Run 2 was too hot — the model was destroyed within ~50 updates; Q *did* eventually move (effective rank collapsed 1708→492) but *after* the model was already behaviourally dead, giving negative lead time. Run 3 split the difference on learning rate and still collapsed, this time caught early by the abort guard at step 110. Three learning rates spanning 2×, three collapses: this now looks structural to KL-free GRPO on this recipe, not like a hyperparameter we keep narrowly missing.

Two secondary observations matter for the writeup. First, the plasticity metrics **lagged** the dashboard in every run where anything happened — reward, clip-ratio and zero-gradient signals fired well before effective rank moved. Second, the length explosion is the likely proximate cause of collapse: in runs 2 and 3, reward fell exactly as completions ballooned to the token cap (99–100% clipped), starving GRPO of gradient. Neither observation supports the detector-beats-dashboard claim as currently framed.

Entropy was the one early hint in run 2 (it spiked before collapse), but run 3's surviving checkpoints show the entropy→learnability relationship running the *opposite* direction. So even the entropy story is regime-dependent and unstable across these three points — worth analysing formally, but not yet a reliable predictor.

Where this leaves the direction

Not a kill, but a required redesign before more compute. The missing ingredient in all three runs is a KL anchor — the proposal's own stability mechanism. $\beta = 0$ is the danger zone, but here it is *too* dangerous: it destroys the model before slow capacity loss can be observed. A small KL ($\beta \approx 0.02$ – 0.04), plus a length control in the reward, is the most likely way to produce a stage A that stays healthy long enough for quiet erosion to appear and be predicted. That is run 4.

There is also a genuine finding to bank regardless: prolonged KL-free RLVR on a small model **reliably self-destructs via length explosion and reward collapse**. That connects directly to the ProRL thread in the related-work section and may be worth reporting in its own right. Recommended next step: take these three runs to the group, decide between (a) redesigning stage A with a KL anchor to chase the detector question, or (b) treating the robust collapse itself as the more interesting result — before spending another session.

Caveat: single seed per run, $\pm 5pp$ eval noise, one model, one stage transition. These are diagnostic pilots, not evidence for or against the proposal's headline claim, which needs the full multi-seed sweep.